

EXHIBIT 3



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Breil et al.

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(54) **SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY CONTACTS**

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(51) **Int. Cl.**

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H01L 31/062 (2012.01)
H01L 31/113 (2006.01)
H01L 31/119 (2006.01)
H01L 29/45 (2006.01)
H01L 29/78 (2006.01)
H01L 29/16 (2006.01)
H01L 29/66 (2006.01)
H01L 29/20 (2006.01)
H01L 21/285 (2006.01)

(52) **U.S. Cl.**

CPC **H01L 29/456** (2013.01); **H01L 21/2855** (2013.01); **H01L 21/28518** (2013.01); **H01L 21/28568** (2013.01); **H01L 21/28575** (2013.01); **H01L 29/16** (2013.01); **H01L 29/20** (2013.01); **H01L 29/452** (2013.01); **H01L 29/665** (2013.01); **H01L 29/66522** (2013.01); **H01L 29/66545** (2013.01); **H01L 29/78** (2013.01)

(58) **Field of Classification Search**

CPC **H01L 21/28518**; **H01L 21/8238**;

H01L 21/823814; **H01L 27/092**; **H01L 29/49**;
H01L 29/4933; **H01L 29/665**; **H01L 29/94**
See application file for complete search history.

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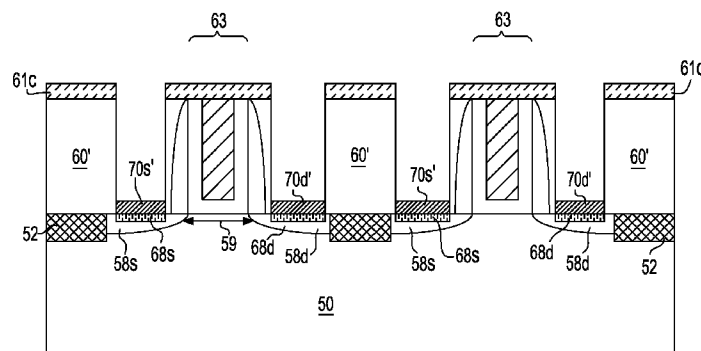
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(57) **ABSTRACT**

Metal semiconductor alloy contacts are provided on each of a source region and a drain region which are present in a semiconductor substrate. A transition metal is then deposited on each of the metal semiconductor alloy contacts, and during the deposition of the transition metal, the deposited transition metal reacts preferably, but not necessarily always, in-situ with a portion of each the metal semiconductor alloy contacts forming a transition metal-metal semiconductor alloy liner atop each metal semiconductor alloy contact. Each transition metal-metal semiconductor alloy liner that is provided has outer edges that are vertically coincident with outer edges of each metal semiconductor alloy contact. The transition metal-metal semiconductor alloy liner is more etch resistant as compared to the underlying metal semiconductor alloy. As such, the transition metal-metal semiconductor alloy liner can serve as an effective etch stop layer during any subsequently performed etch process.

9 Claims, 12 Drawing Sheets



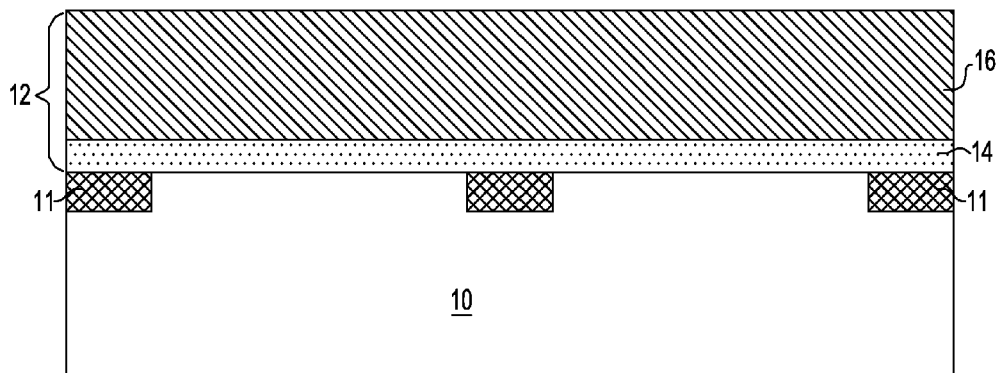


FIG. 1

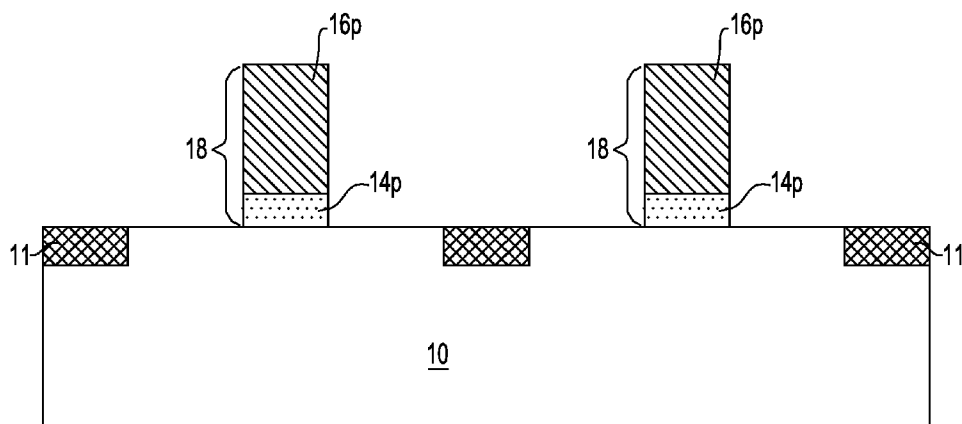


FIG. 2

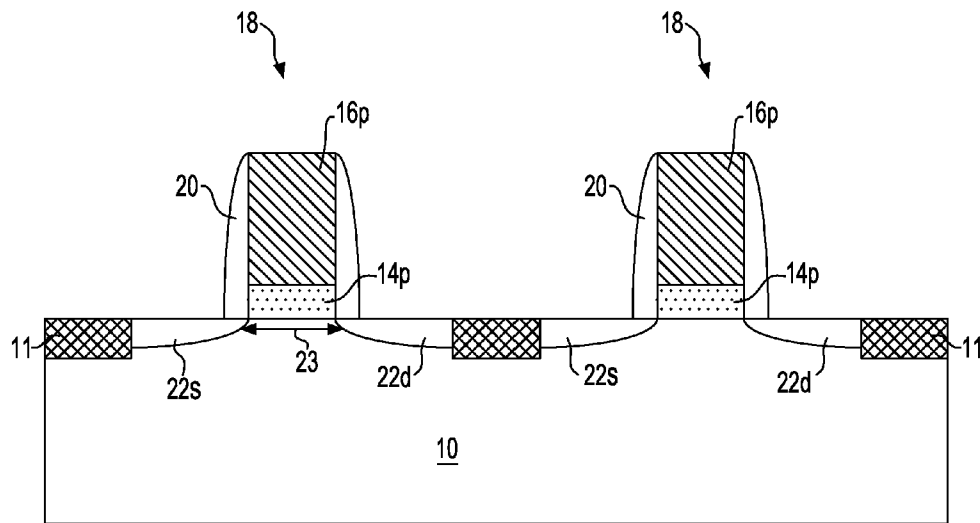


FIG. 3

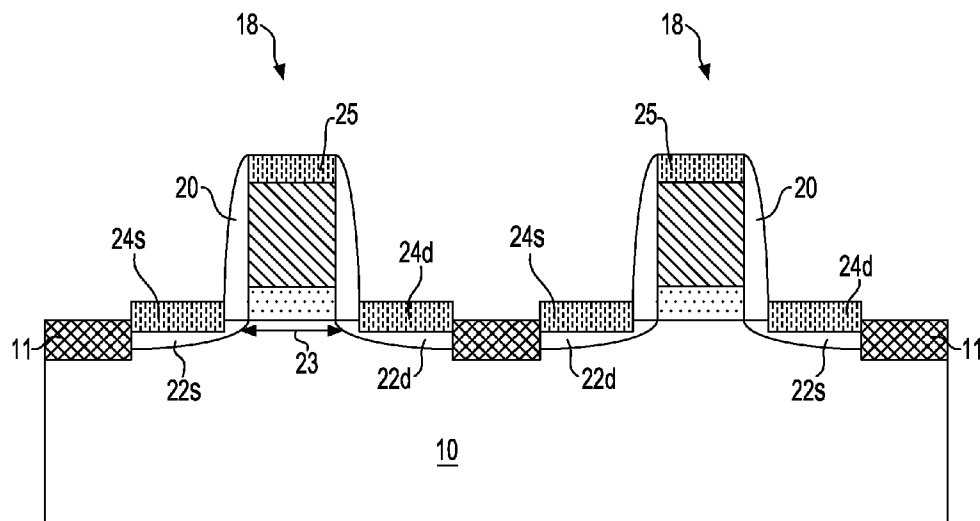


FIG. 4

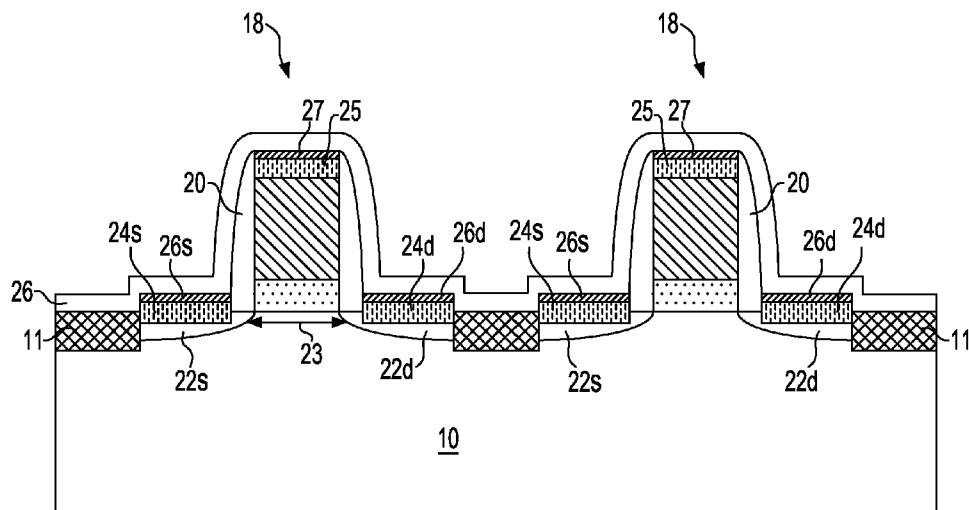


FIG. 5

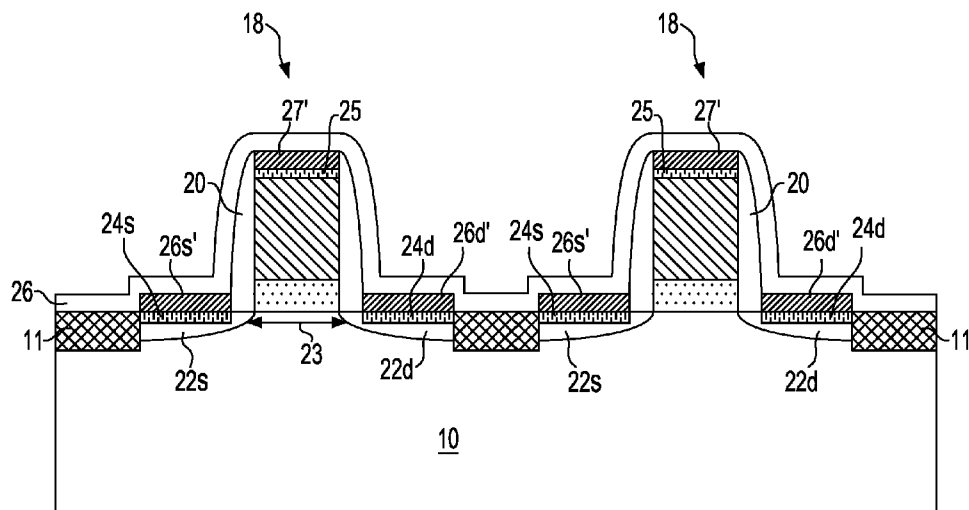


FIG. 6

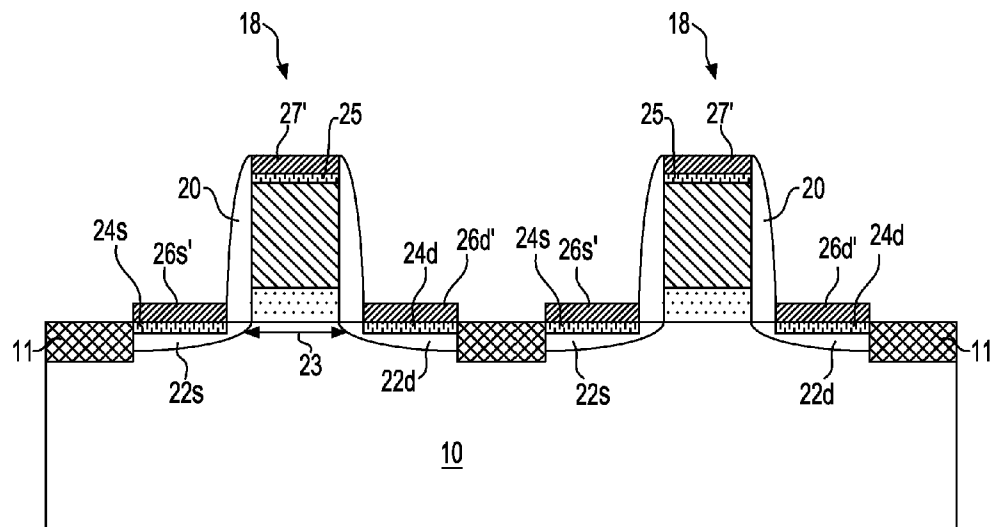


FIG. 7

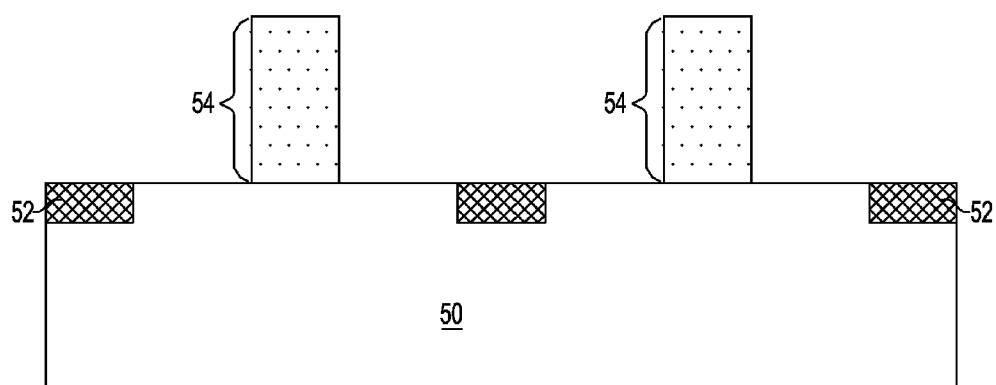


FIG. 8

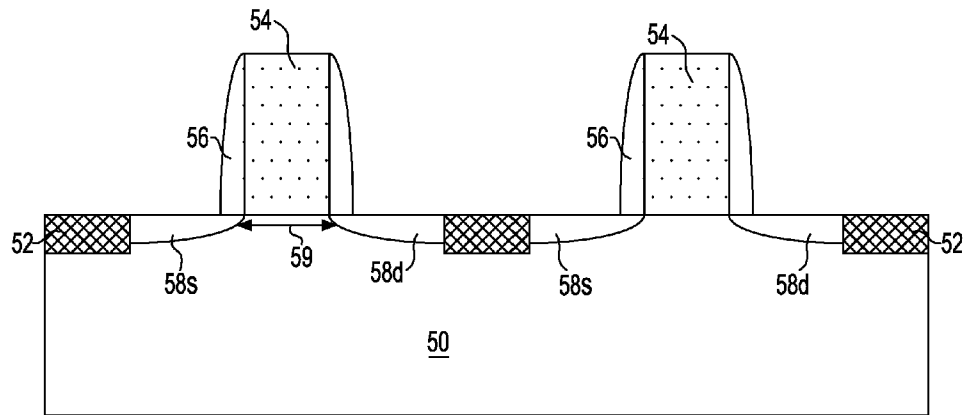


FIG. 9

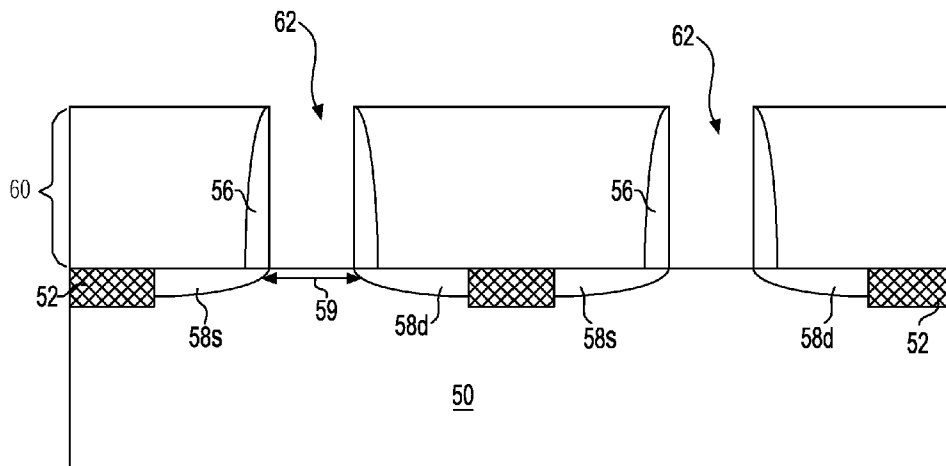


FIG. 10

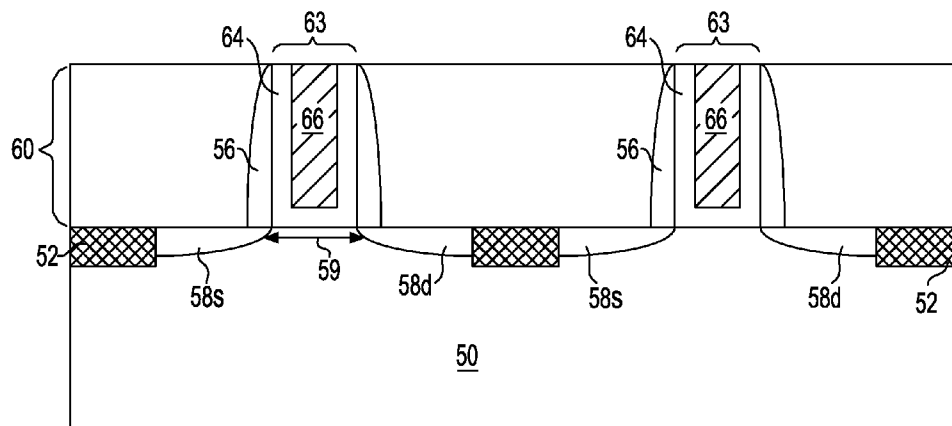


FIG. 11

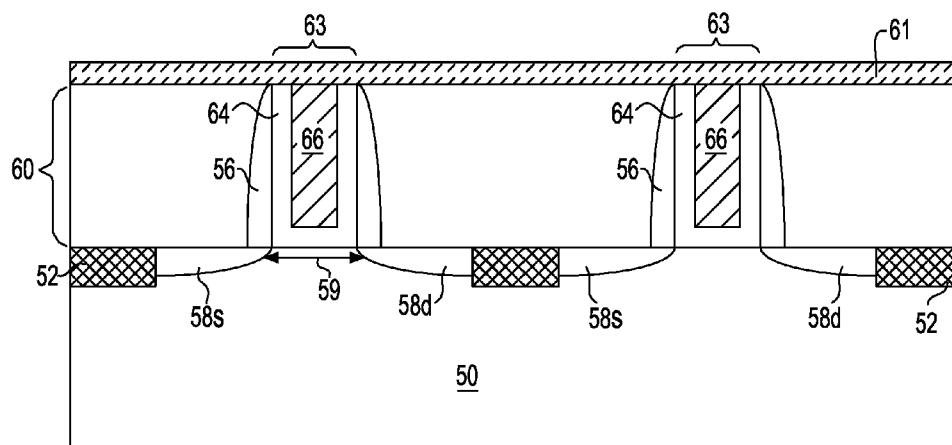


FIG. 12

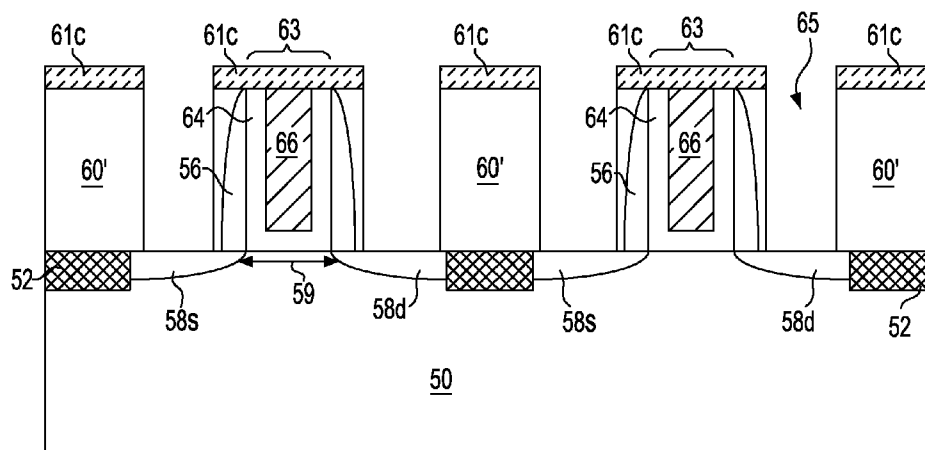


FIG. 13

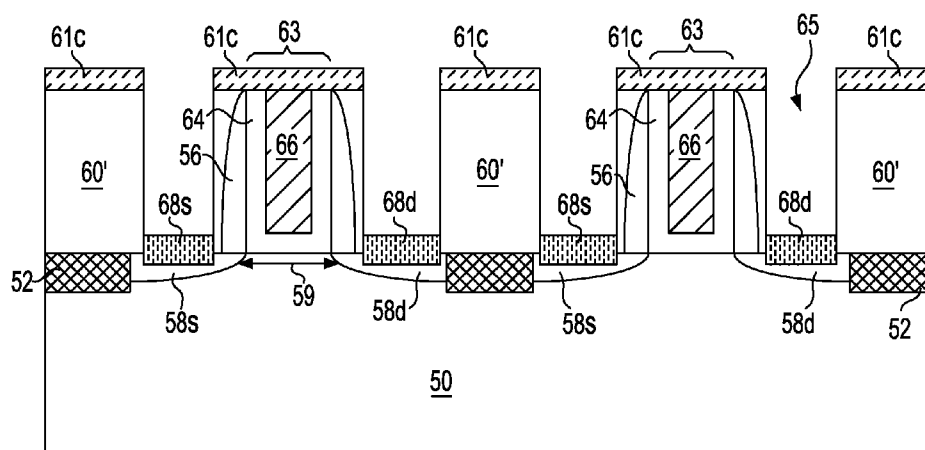


FIG. 14

FIG. 16

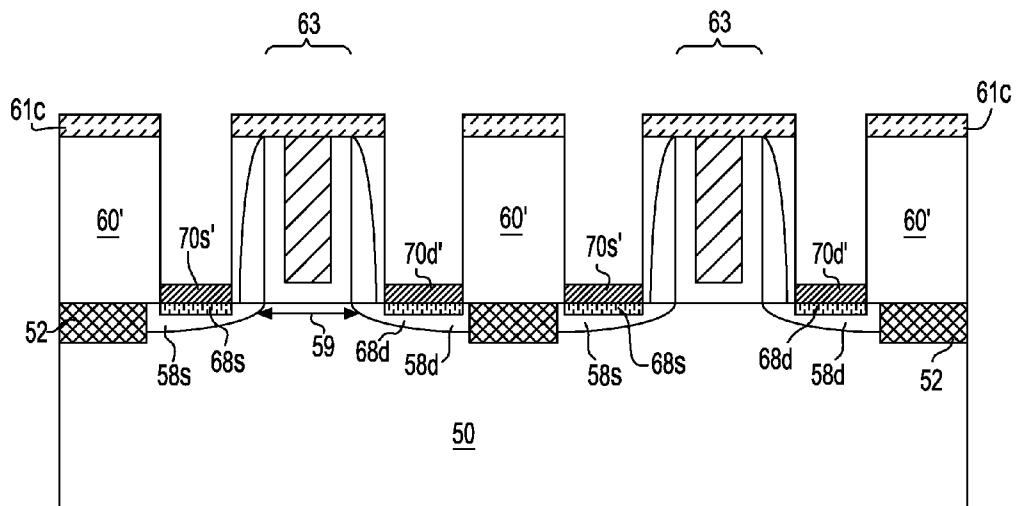


FIG. 17

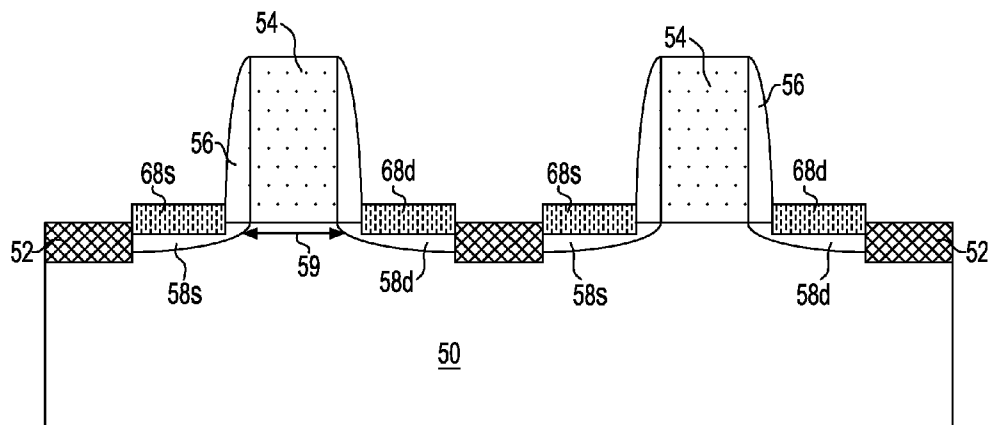


FIG. 18

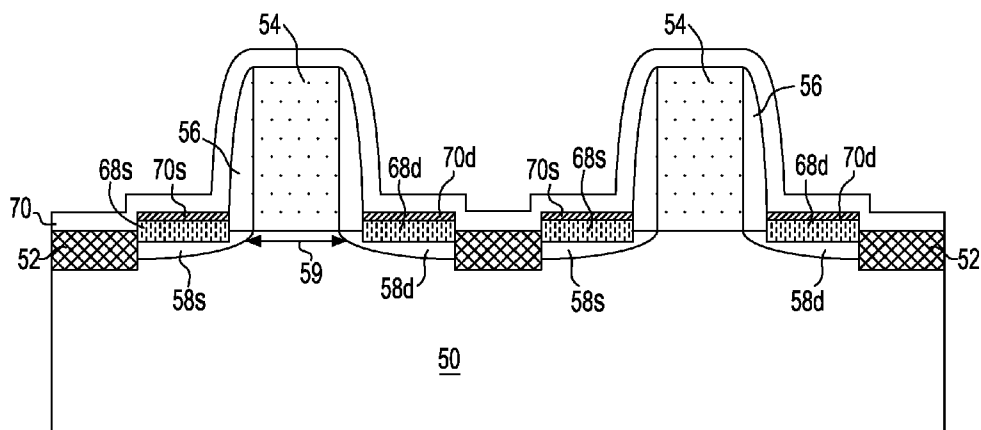


FIG. 19

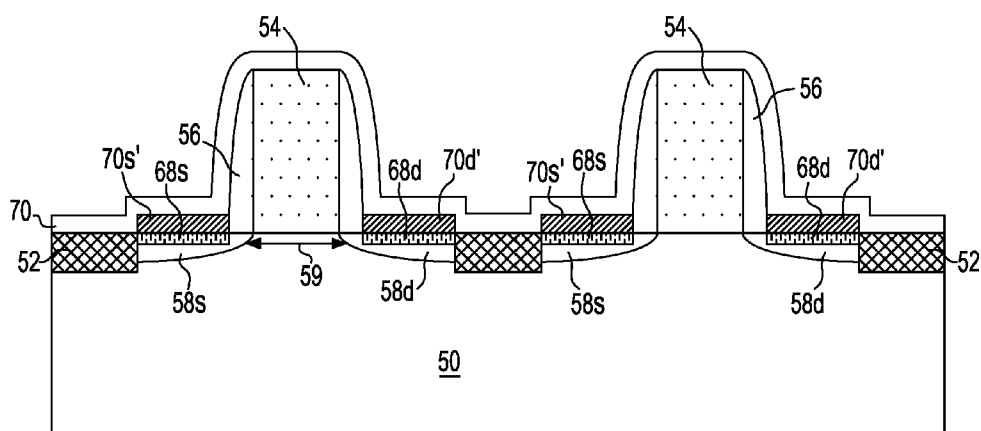


FIG. 20

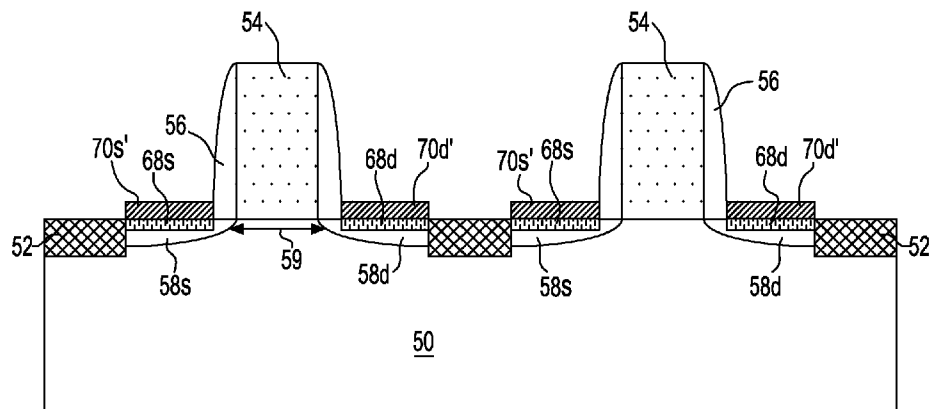


FIG. 21

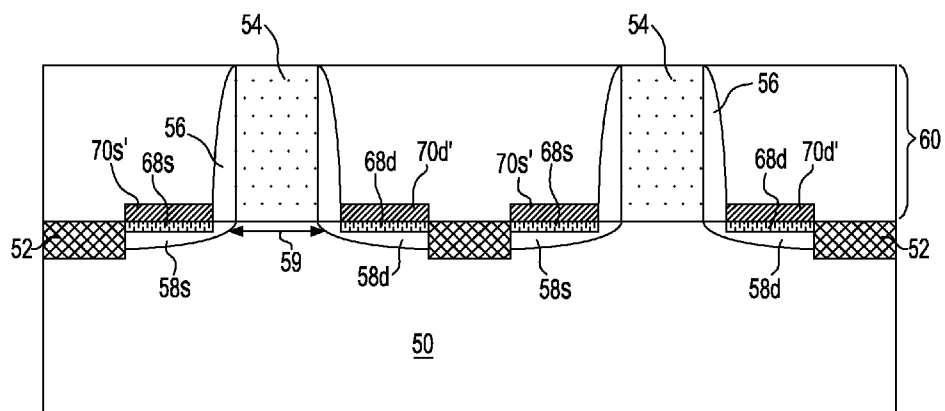


FIG. 22

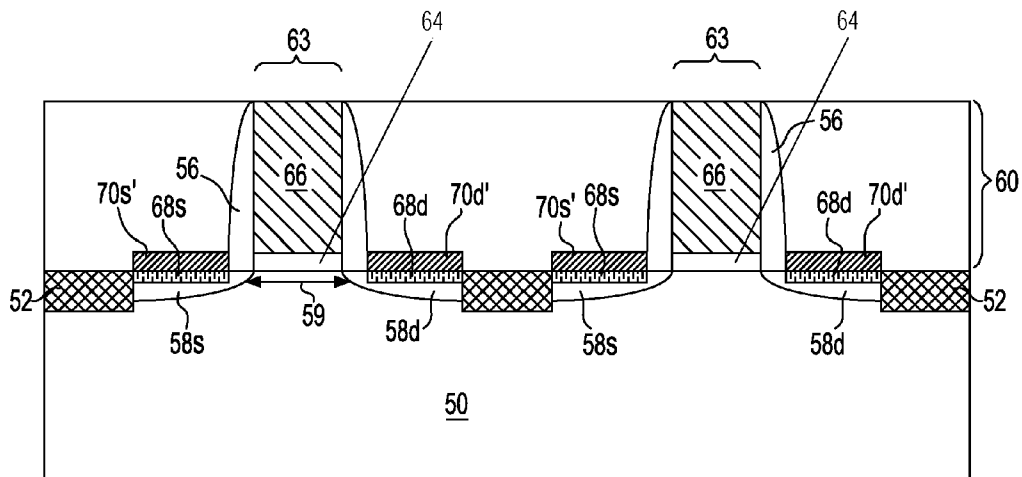


FIG. 23

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SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY CONTACTS

BACKGROUND

The present application relates to semiconductor structures and methods of forming the same. More particularly, the present application relates to semiconductor structures including a stack of, from bottom to top, a metal semiconductor alloy portion and a transition metal-metal semiconductor alloy portion located on each side of a functional gate structure. The present application also provides methods for forming such semiconductor structures.

Field effect transistors (FETs) are the basic building block of today's integrated circuits. Such transistors can be formed in conventional bulk substrates (such as silicon) or in semiconductor-on-insulator (SOI) substrates.

State of the art FETs can be fabricated by depositing a gate conductor over a gate dielectric and a semiconductor substrate. Generally, the FET fabrication process implements lithography and etching processes to define the gate structures. After providing the gate structures, source/drain extensions are formed into a portion of the semiconductor substrate and on both sides of each gate structure by ion implantation. Sometimes this implant is performed using a spacer to create a specific distance between the gate structure and the implanted junction. In some instances, such as in the manufacture of an n-FET device, the source/drain extensions for the n-FET device are implanted with no spacer. For a p-FET device, the source/drain extensions are typically implanted with a spacer present. A thicker spacer is typically formed after the source/drain extensions have been implanted. In some instances, deep source/drain implants can be performed with the thick spacer present. In other instances, and for advanced technologies, the source region and the drain region can be formed using a selective epitaxial growth process. High temperature anneals can be performed to activate the junctions after which the source/drain and top portion of the gate are generally converted into a metal semiconductor alloy (i.e., a metal silicide). The formation of the metal semiconductor alloy typically requires that a transition metal be deposited on the semiconductor substrate followed by a process to produce the metal semiconductor alloy. Such a process forms low resistivity metal semiconductor alloy contacts to the deep source/drain regions.

In current technologies, and among other limiting factors, the metal semiconductor alloy containing contacts can be affected by the following two defects: (1) Encroachment of the metal semiconductor alloy which typically increases leakage through the source/drain junction and can also completely short the source region and the drain region of the device. (2) Spotty (non-uniform) metal semiconductor alloys which may cause unwanted etching of the metal semiconductor alloy contact during a subsequently performed etch.

These two defects are either formed during a subsequently performed etching process or the defects are enhanced by a subsequently performed etching process. Limiting these effects is critical for upcoming technologies.

SUMMARY

Metal semiconductor alloy contacts are provided on each of a source region and a drain region which are present in a semiconductor substrate. A transition metal is then deposited on each of the metal semiconductor alloy contacts, and during the deposition of the transition metal, the deposited transition metal reacts preferably, but not necessarily always, in-situ

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with a portion of each of the metal semiconductor alloy contacts forming a transition metal-metal semiconductor alloy liner atop each metal semiconductor alloy contact. Each transition metal-metal semiconductor alloy liner that is provided has outer edges that are vertically coincident, i.e., vertically aligned, with outer edges of each metal semiconductor alloy contact. The transition metal-metal semiconductor alloy liner is more etch resistant as compared to the underlying metal semiconductor alloy contact. As such, the transition metal-metal semiconductor alloy liner can serve as an effective etch stop layer during any subsequently performed etch process.

In one aspect of the present application, semiconductor structures are provided. The semiconductor structures of the present application include a semiconductor substrate having a source region and a drain region located within a semiconductor material portion of the semiconductor substrate, wherein the source region and the drain region are spaced apart by a channel region. The semiconductor structure also includes a functional gate structure located above the channel region, a source-side material stack of, from bottom to top, a source-side metal semiconductor alloy portion and a source-side transition metal-metal semiconductor alloy portion located on one side of the functional gate structure and located atop the source region, and a drain-side material stack of, from bottom to top, a drain-side metal semiconductor alloy portion and a drain-side transition metal-metal semiconductor alloy portion located on another side of the functional gate structure and located atop the drain region.

In another aspect of the present application, methods of forming semiconductor structures are provided. In one embodiment, the method includes providing a source region and a drain region at a footprint of a functional gate structure. Next a source-side metal semiconductor alloy portion is formed on the source region, and a drain-side metal semiconductor alloy portion is formed on the drain region. A transition metal is then deposited on at least the source-side metal semiconductor alloy portion and the drain-side metal semiconductor alloy portion, wherein during the depositing the transition metal reacts preferably in-situ with the source-side metal semiconductor alloy portion forming a source-side transition metal-metal semiconductor alloy portion on a surface of the source-side metal semiconductor alloy portion, and the transition metal reacts preferably in-situ with the drain-side metal semiconductor alloy portion forming a drain-side transition metal-metal semiconductor alloy portion on a surface of the drain-side metal semiconductor alloy portion. Next, remaining portions of the transition metal are removed.

In another embodiment, the method includes providing a source region and a drain region at a footprint of a sacrificial gate structure. Next, a source-side metal semiconductor alloy portion is formed on the source region, and a drain-side metal semiconductor alloy portion is formed on the drain region. A transition metal is then deposited on at least the source-side metal semiconductor alloy portion and the drain-side metal semiconductor alloy portion, wherein during the depositing the transition metal reacts preferably in-situ with the source-side metal semiconductor alloy portion forming a source-side transition metal-metal semiconductor alloy portion on a surface of the source-side metal semiconductor alloy portion, and the transition metal reacts preferably in-situ with the drain-side metal semiconductor alloy portion forming a drain-side transition metal-metal semiconductor alloy portion on a surface of the drain-side metal semiconductor alloy portion. Next, remaining portions of the transition metal are removed. In accordance with the present application, the sacrificial gate structure can be replaced with a functional gate

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structure prior to forming the source-side metal semiconductor alloy portion on the source region, and the drain-side metal semiconductor alloy portion on the drain region, or after removing the transition metal.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a cross sectional view of a first exemplary semiconductor structure including a material stack located on an upper surface of semiconductor substrate in accordance with an embodiment of the present application.

FIG. 2 is a cross sectional view of the first exemplary semiconductor structure of FIG. 1 after patterning the material stack into at least one functional gate structure.

FIG. 3 is a cross sectional view of the first exemplary semiconductor structure of FIG. 2 after forming a spacer on each vertical sidewall of the at least one functional gate structure and forming a source region in a semiconductor material portion of the semiconductor substrate on one side of the at least one functional gate structure and forming a drain region in another semiconductor material portion of the semiconductor substrate on another side of the at least one functional gate structure.

FIG. 4 is a cross sectional view of the first exemplary semiconductor structure of FIG. 3 after forming a source-side metal semiconductor alloy portion on the source region and a drain-side metal semiconductor alloy portion on the drain region.

FIG. 5 is a cross sectional view of the first exemplary semiconductor structure of FIG. 4 after deposition of a transition metal in which a source-side transition metal-metal semiconductor alloy portion forms preferably in-situ on the source-side metal semiconductor alloy portion and a drain-side transition metal-metal semiconductor alloy portion forms preferably in-situ on the drain-side metal semiconductor alloy portion.

FIG. 6 is a cross sectional view of the first exemplary semiconductor structure of FIG. 5 after annealing.

FIG. 7 is a cross sectional view of the first exemplary semiconductor structure of FIG. 6 after removing remaining portions of the transition metal.

FIG. 8 is a cross sectional view of a second exemplary semiconductor structure including at least one sacrificial gate structure formed on an upper surface of a semiconductor substrate in accordance with another embodiment of the present application.

FIG. 9 is a cross sectional view of the second exemplary semiconductor structure of FIG. 8 after forming a spacer on each vertical sidewall of the at least one sacrificial gate structure and forming a source region on one side of the at least one sacrificial gate structure and a drain region on another side of the at least one sacrificial gate structure.

FIG. 10 is a cross sectional view of the second exemplary semiconductor structure of FIG. 9 after forming a dielectric material and thereafter forming a gate cavity by removing the at least one sacrificial gate structure.

FIG. 11 is a cross sectional view of the second exemplary semiconductor structure of FIG. 10 after forming a gate dielectric portion and a gate conductor portion within the gate cavity.

FIG. 12 is a cross sectional view of the second exemplary semiconductor structure of FIG. 11 after forming a dielectric cap material on an upper surface of the dielectric material.

FIG. 13 is a cross sectional view of the second exemplary semiconductor structure of FIG. 12 after forming a plurality of openings through the dielectric cap material and the dielectric material exposing the source region and the drain region.

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FIG. 14 is a cross sectional view of the second exemplary semiconductor structure of FIG. 13 after forming a source-side metal semiconductor alloy portion on the source region and a drain-side metal semiconductor alloy portion on the drain region.

FIG. 15 is a cross sectional view of the second exemplary semiconductor structure of FIG. 14 after deposition of a transition metal in which a source-side transition metal-metal semiconductor alloy portion forms preferably in-situ on the source-side metal semiconductor alloy portion and a drain-side transition metal-metal semiconductor alloy portion forms preferably in-situ on the drain-side metal semiconductor alloy portion.

FIG. 16 is a cross sectional view of the second exemplary semiconductor structure of FIG. 15 after annealing.

FIG. 17 is a cross sectional view of the second exemplary semiconductor structure of FIG. 16 after removing remaining portions of the transition metal.

FIG. 18 is a cross sectional view of the second exemplary semiconductor structure of FIG. 9 after forming a source-side metal semiconductor alloy portion on the source region and a drain-side metal semiconductor alloy portion on the drain region in accordance with an alternative embodiment of the present application.

FIG. 19 is a cross sectional view of the second exemplary semiconductor structure of FIG. 18 after deposition of a transition metal in which a source-side transition metal-metal semiconductor alloy portion forms preferably in-situ on the source-side metal semiconductor alloy portion and a drain-side transition metal-metal semiconductor alloy portion forms preferably in-situ on the drain-side metal semiconductor alloy portion.

FIG. 20 is a cross sectional view of the second exemplary semiconductor structure of FIG. 19 after annealing.

FIG. 21 is a cross sectional view of the second exemplary semiconductor structure of FIG. 20 after removing remaining portions of the transition metal.

FIG. 22 is a cross sectional view of the second exemplary semiconductor structure of FIG. 21 after forming a dielectric material.

FIG. 23 is a cross sectional view of the second exemplary semiconductor structure of FIG. 22 after forming a gate cavity by removing the at least one sacrificial gate structure, and forming a gate dielectric portion and a gate conductor portion within the gate cavity.

DETAILED DESCRIPTION

The present application will now be described in greater detail by referring to the following discussion and drawings that accompany the present application. It is noted that the drawings of the present application are provided for illustrative purposes only and, as such, the drawings are not drawn to scale. It is also noted that like and corresponding elements are referred to by like reference numerals.

In the following description, numerous specific details are set forth, such as particular structures, components, materials, dimensions, processing steps and techniques, in order to provide an understanding of the various embodiments of the present application. However, it will be appreciated by one of ordinary skill in the art that the various embodiments of the present application may be practiced without these specific details. In other instances, well-known structures or processing steps have not been described in detail in order to avoid obscuring the present application.

The present application can begin by first providing metal semiconductor alloy contacts on each of a source region and

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a drain region which are present in a semiconductor substrate. A transition metal can then be deposited atop each of the metal semiconductor alloy contacts. During the deposition of the transition metal, the deposited transition metal reacts preferably, but not necessarily always, in-situ with a portion of each of the metal semiconductor alloy contacts forming a transition metal-metal semiconductor alloy liner atop each metal semiconductor alloy contact. An optional anneal can be used to thicken the previously formed transition metal-metal semiconductor alloy liner. The transition metal-metal semiconductor alloy liner is more etch resistant as compared to the underlying metal semiconductor alloy contact. As such, the transition metal-metal semiconductor alloy liner can serve as an effective etch stop layer during any subsequently performed etch process. Some specific examples of the formation of the aforementioned transition metal-metal semiconductor alloy liner atop a metal semiconductor alloy contact are now provided.

Referring first to FIG. 1, there is illustrated a first exemplary semiconductor structure including a material stack **12** located on an upper surface of semiconductor substrate **10** in accordance with an embodiment of the present application. The material stack **12** includes, from bottom to top, a gate dielectric material **14** and a gate conductor **16**.

In some embodiments of the present application, the semiconductor substrate **10** can be a bulk semiconductor substrate. When a bulk semiconductor substrate is employed as semiconductor substrate **10**, the bulk semiconductor substrate can be comprised of any semiconductor material including, but not limited to, Si, Ge, SiGe, SiC, SiGeC, and III/V compound semiconductors such as, for example, InAs, GaAs, and InP. Multilayers of these semiconductor materials can also be used as the semiconductor material of the bulk semiconductor. In one embodiment, the semiconductor substrate **10** can be comprised of a single crystalline semiconductor material, such as, for example, single crystalline silicon. In other embodiments, the semiconductor substrate **10** may comprise a polycrystalline or amorphous semiconductor material.

In another embodiment, a semiconductor-on-insulator (SOI) substrate (not specifically shown) can be employed as the semiconductor substrate **10**. Although not specifically shown, one skilled in the art understands that an SOI substrate includes a handle substrate, a buried insulator layer located on an upper surface of the handle substrate, and a semiconductor layer located on an upper surface of the buried insulator layer. The handle substrate provides mechanical support for the buried insulator layer and the semiconductor layer.

The handle substrate and the semiconductor layer of the SOI substrate may comprise the same, or different, semiconductor material. The term "semiconductor" as used herein in connection with the semiconductor material of the handle substrate and the semiconductor layer denotes any semiconductor material including, for example, Si, Ge, SiGe, SiC, SiGeC, and III/V compound semiconductors such as, for example, InAs, GaAs, or InP. Multilayers of these semiconductor materials can also be used as the semiconductor material of the handle substrate and the semiconductor layer. In one embodiment, the handle substrate and the semiconductor layer are both comprised of silicon. In some embodiments, the handle substrate is a non-semiconductor material including, for example, a dielectric material and/or a conductive material. In yet other embodiments, the handle substrate can be omitted and a substrate including an insulator layer and a semiconductor layer can be used as semiconductor substrate **10**.

In some embodiments, the handle substrate and the semiconductor layer may have the same or different crystal orientation.

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For example, the crystal orientation of the handle substrate and/or the semiconductor layer may be {100}, {110}, or {111}. Other crystallographic orientations besides those specifically mentioned can also be used in the present application. The handle substrate and/or the semiconductor layer of the SOI substrate may be a single crystalline semiconductor material, a polycrystalline material, or an amorphous material. Typically, at least the semiconductor layer is a single crystalline semiconductor material. In some embodiments, the semiconductor layer that is located atop the buried insulator layer can be processed to include semiconductor regions having different crystal orientations.

The buried insulator layer of the SOI substrate may be a crystalline or non-crystalline oxide or nitride. In one embodiment, the buried insulator layer is an oxide such as, for example, silicon dioxide. The buried insulator layer may be continuous or it may be discontinuous. When a discontinuous buried insulator region is present, the insulator region exists as an isolated island that is surrounded by semiconductor material.

The SOI substrate may be formed utilizing standard processes including for example, SIMOX (separation by ion implantation of oxygen) or layer transfer. When a layer transfer process is employed, an optional thinning step may follow the bonding of two semiconductor wafers together. The optional thinning step reduces the thickness of the semiconductor layer to a layer having a thickness that is more desirable.

In one example, the thickness of the semiconductor layer of the SOI substrate can be from 100 Å to 1000 Å. In another example, the thickness of the semiconductor layer of the SOI substrate can be from 500 Å to 700 Å. In some embodiments, and when an ETSOI (extremely thin semiconductor-on-insulator) substrate is employed, the semiconductor layer of the SOI has a thickness of less than 100 Å. If the thickness of the semiconductor layer is not within one of the above mentioned ranges, a thinning step such as, for example, planarization or etching can be used to reduce the thickness of the semiconductor layer to a value within one of the ranges mentioned above. The buried insulator layer of the SOI substrate typically has a thickness from 10 Å to 2000 Å, with a thickness from 1000 Å to 1500 Å being more typical. The thickness of the handle substrate of the SOI substrate is inconsequential to the present application.

In some other embodiments, hybrid semiconductor substrates which have different surface regions of different crystallographic orientations can be employed as semiconductor substrate **10**. When a hybrid substrate is employed, an nFET is typically formed on a (100) crystal surface, while a pFET is typically formed on a (110) crystal plane. The hybrid substrate can be formed by techniques that are well known in the art. See, for example, U.S. Pat. No. 7,329,923, U.S. Publication No. 2005/0116290, dated Jun. 2, 2005 and U.S. Pat. No. 7,023,055, the entire contents of each are incorporated herein by reference.

Semiconductor substrate **10** may be doped, undoped or contain doped and undoped regions therein. For clarity, the doped regions are not specifically shown in the drawings of the present application. Each doped region within the semiconductor material of the semiconductor substrate **10** may have the same, or they may have different conductivities and/or doping concentrations. The doped regions that are present in the semiconductor material of semiconductor substrate **10** are typically referred to as well regions and they are formed utilizing a conventional ion implantation process, gas phase doping or epitaxial growth.

In some embodiments, the semiconductor substrate **10** can be processed to include at least one isolation region **11** therein. The at least one isolation region **11** can be a trench isolation region or a field oxide isolation region. The trench isolation region can be formed utilizing a conventional trench isolation process well known to those skilled in the art. For example, lithography, etching and filling of the trench with a trench dielectric such as an oxide may be used in forming the trench isolation region. Optionally, a liner may be formed in the trench prior to trench fill, a densification step may be performed after the trench fill and a planarization process may follow the trench fill as well. The field oxide isolation region may be formed utilizing a so-called local oxidation of silicon process. Note that the at least one isolation region **11** may provide isolation between neighboring gate structures, typically required when the neighboring gate structures have opposite conductivities, i.e., n-type transistors and p-type transistors. As such, the at least one isolation region **11** can separate a first device region in which an n-type transistor device can be formed and second device region in which a p-type transistor device can be formed.

A material stack **12** is formed on an upper surface of the semiconductor substrate **10** and on an upper surface of each isolation region **11**. As stated above, the material stack **12** includes, from bottom to top, a gate dielectric material **14** and a gate conductor **16**. The gate dielectric material **14** and the gate conductor **16** which provide material stack **12** are formed as blanket layers atop the upper surface of the semiconductor substrate **10**.

In this embodiment of the present application, the material stack **12** is used in providing a functional gate structure. The term "functional gate structure" is used throughout the present application as a permanent gate structure used to control output current (i.e., flow of carriers in the channel) of a semiconducting device through electrical or magnetic fields.

The gate dielectric material **14** that can be used in the present application can be a semiconductor oxide, a semiconductor nitride, and/or a semiconductor oxynitride. In one example, the gate dielectric material **14** can be composed of silicon dioxide, silicon nitride and/or silicon oxynitride. In another embodiment of the present application, the gate dielectric material **14** may include at least a dielectric metal oxide. Exemplary dielectric metal oxides that can be used as gate dielectric material **14** include, but are not limited to, HfO_2 , ZrO_2 , La_2O_3 , Al_2O_3 , TiO_2 , SrTiO_3 , LaAlO_3 , Y_2O_3 , HfO_xN_y , ZrO_xN_y , $\text{La}_2\text{O}_x\text{N}_y$, $\text{Al}_2\text{O}_x\text{N}_y$, TiO_xN_y , SrTiO_xN_y , LaAlO_xN_y , $\text{Y}_2\text{O}_x\text{N}_y$, SiON , SiN_x , a silicate thereof, and an alloy thereof. Each value of x is independently from 0.5 to 3 and each value of y is independently from 0 to 2. In some embodiments, a multilayered gate dielectric structure comprising different gate dielectric materials, e.g., silicon dioxide, and a dielectric metal oxide can be formed and used as the gate dielectric material **14**.

In some embodiments of the present application, the gate dielectric material **14** can be formed by a deposition technique such as, for example, chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), physical vapor deposition (PVD), sputtering, or atomic layer deposition. In another embodiment of the present application, the gate dielectric material **14** can be formed by a thermal growth technique such as, for example, thermal oxidation and/or thermal nitridation. In yet a further embodiment of the present application, a combination of a deposition and thermal growth may be used in forming a multilayered gate dielectric structure.

In one embodiment of the present application, the gate dielectric material **14** can have a thickness in a range from 1 nm to 10 nm. Other thicknesses that are lesser than or greater than the aforementioned thickness range can also be employed for the gate dielectric material **14**.

The gate conductor **16** that can be used in the present application can be composed of doped polysilicon, doped silicon germanium, an elemental metal (e.g., tungsten, titanium, tantalum, aluminum, nickel, ruthenium, palladium and platinum), an alloy of at least two elemental metals, an elemental metal nitride (e.g., tungsten nitride, aluminum nitride, and titanium nitride), or multilayered combinations thereof. In one embodiment, an entirety of the gate conductor **16** is comprised of a doped polysilicon or doped polysilicon germanium. In another embodiment, a lower portion of the gate conductor **16** is comprised of a conductive material other than doped polysilicon or doped polysilicon germanium, and an upper portion of the gate conductor **16** is comprised of doped polysilicon or doped silicon germanium.

The gate conductor **16** can be formed utilizing a deposition process including, for example, chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), physical vapor deposition (PVD), sputtering, atomic layer deposition (ALD) or other like deposition processes. In one embodiment, the gate conductor **16** has a thickness from 1 nm to 100 nm. Other thicknesses that are lesser than or greater than the aforementioned thickness range can also be employed for the gate conductor **16**.

Referring now to FIG. 2, there is illustrated the first exemplary semiconductor structure of FIG. 1 after patterning the material stack **12** (including gate dielectric material **14** and gate conductor **16**) into at least one functional gate structure **18**. The at least one functional gate structure **18** includes, from bottom to top, a gate dielectric portion **14p** and a gate electrode portion **16p**. Gate dielectric portion **14p** includes a remaining portion of the gate dielectric material **14** of material stack **12**, while gate conductor portion **16p** includes a remaining portion of the gate conductor **16** of material stack **12**.

The at least one functional gate structure **18** can be formed by lithography and etching. Lithography can include forming a photoresist (not shown) on the topmost surface of gate conductor **16** exposing the photoresist to a desired pattern of radiation, and then developing the exposed photoresist with a resist developer to provide a patterned photoresist atop the gate conductor **16** of material stack **12**. At least one etch is then employed which transfers the pattern from the patterned photoresist into the various materials. In one embodiment, the etch used for pattern transfer may include a dry etch process such as, for example, reactive ion etching, plasma etching, ion beam etching or laser ablation. In another embodiment, the etch used for pattern transfer may include a wet chemical etchant such as, for example, KOH (potassium hydroxide). In yet another embodiment, a combination of a dry etch and a wet chemical etch may be used to transfer the pattern. After transferring the pattern into the material layers, the patterned photoresist can be removed utilizing a resist stripping process such as, for example, ashing.

As is shown in the embodiment illustrated in FIG. 2, the sidewall edges of the gate dielectric material portion **14p** and the gate conductor **16p** are vertically coincident to (i.e., vertically aligned with) each other.

Referring now to FIG. 3, there is illustrated the first exemplary semiconductor structure of FIG. 2 after forming a spacer **20** on each vertical sidewall of the at least one functional gate structure **18** and forming a source region **22s** in a semiconductor material portion of the semiconductor sub-

strate **10** on one side of the at least one functional gate structure **18** and forming a drain region **22d** in another semiconductor material portion of the semiconductor substrate **10** on another side of the at least one functional gate structure **18**. The semiconductor portion of the semiconductor substrate **10** that is located beneath the functional gate structure **18** and positioned between the source region **22s** and the drain region **22d** is referred as a channel region **23**.

The spacer **20** can be formed by first providing a spacer material and then etching the spacer material. The spacer material may be composed of any dielectric spacer material including, for example, a dielectric oxide, dielectric nitride, and/or dielectric oxynitride. In one example, the spacer material used in providing the spacer **20** may be composed of silicon dioxide or silicon nitride. The spacer material can be provided by a deposition process including, for example, chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), or physical vapor deposition (PVD). The etching of the spacer material may comprise a dry etch process such as, for example, a reactive ion etch.

FIG. 3 also shows the presence of a source region **22s** located within a semiconductor material portion of the semiconductor substrate **10** on one side of the at least one functional gate structure **18** and a drain region **22d** located within another semiconductor material portion of the semiconductor substrate **10** on another side of the at least one functional gate structure **18**. In some embodiments, a source extension region and drain extension region (not separately shown) are typically formed prior to forming the spacer **20** utilizing an extension ion implantation process. As such, a portion of each of the source extension region and drain extension region would be located beneath the spacer **20**. After forming the spacer **20**, source region **22s** and drain region **22d** are formed utilizing a source/drain ion implantation process. An activation anneal may follow the implantation processes.

The source region **22s** (including the corresponding source extension region) and the drain region (including the corresponding drain extension region) may be doped with a p-type or n-type dopant. The term “p-type” refers to the addition of impurities to an intrinsic semiconductor that creates deficiencies of valence electrons. In a silicon-containing substrate, examples of p-type dopants, i.e., impurities, include, but are not limited to, boron, aluminum, gallium and indium. “N-type” refers to the addition of impurities that contributes free electrons to an intrinsic semiconductor. In a silicon containing substrate examples of n-type dopants, i.e., impurities, include, but are not limited to, antimony, arsenic and phosphorous. The concentration of dopants within the source region **22s** (and corresponding source extension region) and the drain region **22d** (and the corresponding drain extension region) can be within ranges typically used in forming metal oxide semiconductor field effect transistors (MOSFETs).

Referring now to FIG. 4, there is illustrated the first exemplary semiconductor structure of FIG. 3 after forming a source-side metal semiconductor alloy portion **24s** on the source region **22s** and a drain-side metal semiconductor alloy portion **24d** on the drain region **22d**. The source-side metal semiconductor alloy portion **24s**, and the drain-side metal semiconductor alloy portion **24d** are comprised of a same metal semiconductor alloy.

In some embodiments such as shown in FIG. 4 and when a semiconductor material is present at an upper surface of the gate conductor portion **18p**, a metal semiconductor alloy cap portion **25** can be formed on the gate conductor portion **18p**. In some embodiments, the source-side metal semiconductor alloy portion **24s**, the drain-side metal semiconductor alloy portion **24d** and optionally the metal semiconductor alloy cap

portion **25** each include a same metal semiconductor alloy. In another embodiment, the source-side metal semiconductor alloy portion **24s**, and the drain-side metal semiconductor alloy portion **24d** each comprise a first metal semiconductor alloy, while the optional metal semiconductor alloy cap portion **25** comprises a second metal semiconductor alloy that is different from the first metal semiconductor alloy. The difference between the first metal semiconductor alloy and the second metal semiconductor alloy is in respect to the semiconductor element, not the metallic element that forms each metal semiconductor alloy.

The source-side metal semiconductor alloy portion **24s**, the drain-side metal semiconductor alloy portion **24d** and optionally the metal semiconductor alloy cap portion **25**, are formed by first providing a layer of a metal semiconductor alloy forming metal (not shown) on all exposed surfaces of the structure shown in FIG. 3. The term “metal semiconductor alloy forming metal” is used throughout the present application to denote a metal that can react with an underlying semiconductor material to form a metal semiconductor alloy. Illustrative examples of metal semiconductor alloy forming metals that can be used in the present application include at least one of nickel (Ni), platinum (Pt), palladium (Pd), titanium (Ti), tungsten (W), and cobalt (Co). The metal semiconductor alloy forming metal can be formed by a deposition process including, for example, chemical vapor deposition, plasma enhanced chemical vapor deposition, plating or sputtering. In some embodiments, a co-deposition of metal semiconductor alloy forming metals can be used. In another embodiment, a first metal semiconductor alloy forming metal can be formed, followed by a second metal semiconductor alloy forming metal. The metal semiconductor alloy metal that is formed can have a thickness from 5 to 15 nm. Other thicknesses that are greater than or lesser than the aforementioned thickness range can also be employed as the thickness of the metal semiconductor alloy forming metal.

After providing the metal semiconductor alloy forming metal and in some embodiments of the present application, a diffusion barrier (not shown) can be formed on an exposed upper surface of the metal semiconductor alloy forming metal. In another embodiment of the present application, no diffusion barrier is provided on the exposed upper surface of the metal semiconductor alloy forming metal. When present, the diffusion barrier can include a metal nitride such as, for example, TiN or TaN, and any deposition process including those mentioned above for providing the metal semiconductor alloy forming metal may be used. When present, the diffusion barrier can have a thickness from 1 nm to 20 nm.

Next, an anneal is performed under conditions that are effective in causing the metal semiconductor alloy forming metal to diffuse into the semiconductor material portions of the source region **22s**, the drain region **22d** and optionally the gate conductor portion **18p**, forming a metal semiconductor alloy. Notably, the anneal can form the source-side metal semiconductor alloy portion **24s**, the drain-side metal semiconductor alloy portion **24d** and optionally the metal semiconductor alloy cap portion **25**. In one embodiment, the source-side metal semiconductor alloy portion **24s**, the drain-side metal semiconductor alloy portion **24d** and optionally the metal semiconductor alloy cap portion **25** each comprise a metal silicide such as, for example, nickel silicide or platinum-nickel silicide. In another embodiment, the source-side metal semiconductor alloy portion **24s**, the drain-side metal semiconductor alloy portion **24d** and optionally the metal semiconductor alloy cap portion **25** each include a metal germanide such as nickel germanide.

The metal semiconductor alloy formation anneal may be performed in a single step or a two-step anneal can be used. In one embodiment and when nickel is used, the metal semiconductor alloy formation anneal can be performed at a temperature of from 200° C. to 500° C. In another embodiment, temperatures greater than 500° C. can be used. The metal semiconductor alloy formation anneal is typically performed in an ambient including, for example, argon, helium, neon and/or nitrogen. The metal semiconductor alloy formation anneal can be performed utilizing a rapid thermal anneal, a spike anneal, a microwave anneal or a laser anneal. Following the metal semiconductor alloy formation anneal, the optional diffusion barrier and any unreacted metal semiconductor alloy forming metal can be removed utilizing one or more etch processes.

The above processing forms a source-side metal semiconductor alloy portion 24s that has outer edges that are vertically aligned with outer edges of the source region 22s, a drain-side metal semiconductor alloy portion 24d that has outer edges that are vertical aligned with outer edges of the drain region 22d, and an optional metal semiconductor alloy cap portion 25 that has outer edges that are vertically aligned with outer edges of a remaining portion of the gate conductor portion 16p. In some embodiments of the present application, the metal semiconductor alloy formation anneal may partially or entirely consume the semiconductor material portion of the source region 22s and the drain region 22d.

Referring now to FIG. 5, there is illustrated the first exemplary semiconductor structure of FIG. 4 after deposition of a transition metal 26 in which a source-side transition metal-metal semiconductor alloy portion 26s forms preferably in-situ on the source-side metal semiconductor alloy portion 24s and a drain-side transition metal-metal semiconductor alloy portion 26d forms preferably in-situ on the drain-side metal semiconductor alloy portion 24d. An optional transition metal-metal semiconductor alloy cap portion 27 can also be formed preferably in-situ on the metal semiconductor alloy cap portion 25. The source-side transition metal-metal semiconductor alloy portion 26s, the drain-side transition metal-metal semiconductor alloy portion 26d, and optionally the transition metal-metal semiconductor alloy cap portion 27 are formed simultaneously with the deposition of transition metal 26 by diffusion and intermixing of the transition metal with the underlying source-side metal semiconductor alloy portion 24s, drain-side metal semiconductor alloy portion 24d, and metal semiconductor alloy cap portion 25, respectively.

The transition metal 26 that provides the source-side transition metal-metal semiconductor alloy portion 26s, the drain-side transition metal-metal semiconductor alloy portion 26d, and optionally the transition metal-metal semiconductor alloy cap portion 27 comprises any transition metal that can diffuse and interact with the underlying source-side metal semiconductor alloy portion 24s, the drain-side metal semiconductor alloy portion 24d and, optionally, the metal-metal semiconductor alloy cap portion 25, with the proviso that the transition metal 26 is different from the metal semiconductor alloy forming metal mentioned above. Examples of transition metals that may be employed at this point of the present application include transition metals from Groups 4, 5 and 6 of the Periodic Table such as, for example, titanium (Ti), zirconium (Zr), hafnium (Hf), tantalum (Ta), tungsten (W), niobium (Nb), molybdenum (Mo), vanadium (V), and chromium (Cr). Rhenium (Re) can also be used as the transition metal 26. In some embodiments in which nFETs and pFETs are formed, different compositions of transition metals can be used in forming the source-side transition metal-metal semi-

conductor alloy portion 26s and the drain-side transition metal-metal semiconductor alloy portion 26d.

The transition metal 26 that provides the source-side transition metal-metal semiconductor alloy portion 26s, the drain-side transition metal-metal semiconductor alloy portion 26d, and optionally the transition metal-metal semiconductor alloy cap portion 27 can be formed by any of the deposition processes mentioned above in forming the metal semiconductor alloy forming metal. In one example, the transition metal 26 that provides the source-side transition metal-metal semiconductor alloy portion 26s, the drain-side transition metal-metal semiconductor alloy portion 26d, and optionally the transition metal-metal semiconductor alloy cap portion 27 can be formed by sputtering. In one embodiment, the sputtering can be performed at a temperature from room temperature (i.e., 20° C.) up to, and including 500° C. The thickness of the transition metal 26 that provides the source-side transition metal-metal semiconductor alloy portion 26s, the drain-side transition metal-metal semiconductor alloy portion 26d, and optionally the transition metal-metal semiconductor alloy cap portion 27 can be from 2 nm to 10 nm. Other thicknesses can also be employed for the thickness of the transition metal 26 that provides the source-side transition metal-metal semiconductor alloy portion 26s, the drain-side transition metal-metal semiconductor alloy portion 26d, and optionally the transition metal-metal semiconductor alloy cap portion 27.

In accordance with the present application, the source-side transition metal-metal semiconductor alloy portion 26s is composed on an intermixed product of the transition metal 26 and the previously formed source-side metal semiconductor alloy portion 24s. Typically, the source-side transition metal-metal semiconductor alloy portion 26s can contain up to 70% of the transition metal 26. In one example, and when the transition metal 26 comprises titanium, and the source-side metal semiconductor alloy portion 24s comprises nickel silicide, then the source-side transition metal-metal semiconductor alloy portion 26s comprises titanium-nickel silicide. In some embodiments, the source-side transition metal-metal semiconductor alloy portion 26s may comprise titanium-platinum-nickel silicide. The source-side transition metal-metal semiconductor alloy portion 26s has outer edges that are vertically aligned with outer edges of the source-side metal semiconductor alloy portion 24s. In one embodiment, the source-side transition metal-metal semiconductor alloy portion 26s has a first thickness from 1 nm to 10 nm. In another embodiment, the source-side transition metal-metal semiconductor alloy portion 26s has a first thickness from 1 nm to 3 nm.

In accordance with the present application, the drain-side transition metal-metal semiconductor alloy portion 26d is composed on an intermixed product of the transition metal 26 and the previously formed drain-side metal semiconductor alloy portion 24d. Typically, the drain-side transition metal-metal semiconductor alloy portion 26d can contain up to 70% of the transition metal 26. In one example, and when the transition metal 26 comprises titanium, and the drain-side metal semiconductor alloy portion 24d comprises nickel silicide, then the drain-side transition metal-metal semiconductor alloy portion 26d comprises titanium-nickel silicide. In some embodiments, the drain-side transition metal-metal semiconductor alloy portion 26d may comprise titanium-platinum-nickel silicide. The drain-side transition metal-metal semiconductor alloy portion 26d has outer edges that are vertically aligned with outer edges of the drain-side metal semiconductor alloy portion 24d. In one embodiment, the drain-side transition metal-metal semiconductor alloy por-

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tion **26d** has a first thickness from 1 nm to 10 nm. In one embodiment, the drain-side transition metal-metal semiconductor alloy portion **26d** has a first thickness from 1 nm to 3 nm. The source-side metal semiconductor-alloy portion **24s** and the drain-side metal semiconductor alloy portion **24d** that remain after forming source-side transition metal-metal semiconductor alloy portion **26s** and the drain-side transition metal-metal semiconductor alloy portion **26d** can have a thickness from 3 nm to 25 nm, with a thickness from 10 nm to 20 nm being more typical.

In accordance with the present application, the transition metal-metal semiconductor alloy cap portion **27** is composed on an intermixed product of the transition metal **26** and the previously formed metal semiconductor alloy cap portion **25**. In one example, and when the transition metal **26** comprises titanium, and the metal semiconductor alloy cap portion **25** comprises nickel silicide, then the transition metal-metal semiconductor alloy cap portion **27** comprises titanium-nickel silicide. In some embodiments, transition metal-metal semiconductor alloy cap portion **27** may comprise titanium-platinum-nickel silicide. The transition metal-metal semiconductor alloy cap portion **27** has outer edges that are vertically aligned with outer edges of the metal semiconductor alloy cap portion **25**. In one embodiment, the transition metal-metal semiconductor alloy cap portion **27** has a first thickness from 1 nm to 10 nm. In one embodiment, the transition metal-metal semiconductor alloy cap portion **27** has a first thickness from 1 nm to 3 nm.

The source-side transition metal-metal semiconductor alloy portion **26s**, the drain-side transition metal-metal semiconductor alloy portion **26d**, and optionally the transition metal-metal semiconductor alloy cap portion **27** have a higher etch resistance (in either a wet etch or a dry etch process) than the source-side metal semiconductor alloy portion **24s**, the drain-side metal semiconductor alloy portion **24d**, and optionally the metal semiconductor alloy cap portion **25**.

It is noted that the remaining portions of the source-side metal semiconductor alloy portion **24s**, the drain-side metal semiconductor alloy portion **24d**, and optionally the metal semiconductor alloy cap portion **25** have a thickness that is less than the thickness of each of the source-side metal semiconductor alloy portion **24s**, the drain-side metal semiconductor alloy portion **24d**, and optionally the metal semiconductor alloy cap portion **25** prior to the formation of the source-side transition metal-metal semiconductor alloy portion **26s**, the drain-side transition metal-metal semiconductor alloy portion **26d**, and optionally the transition metal-metal semiconductor alloy cap portion **27**.

Referring now to FIG. 6, there is illustrated the first exemplary semiconductor structure of FIG. 5 after annealing. This anneal step is optional and need not be performed in all instances. The anneal process serves to thicken the previously formed source-side transition metal-metal semiconductor alloy portion **26s**, the drain-side transition metal-metal semiconductor alloy portion **26d**, and optionally the transition metal-metal semiconductor alloy cap portion **27**. The thickened source-side transition metal-metal semiconductor alloy portion is labeled as element **26s'**, the thickened drain-side transition metal-metal semiconductor alloy portion is labeled as **26d'**, and the optional thickened transition metal-metal semiconductor alloy cap portion is labeled as **27'** in FIG. 6.

Notably, the anneal provides a thickened source-side transition metal-metal semiconductor alloy portion **26s'** having a second thickness that is greater than the first thickness of the previously formed source-side transition metal-metal semiconductor alloy portion **26s**, a thickened drain-side transition metal-metal semiconductor alloy portion **26d'** having a sec-

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ond thickness that is greater than the first thickness of the previously formed drain-side transition metal-metal semiconductor alloy portion **26d**, and, if present, the thickened transition metal-metal semiconductor alloy cap portion **27'** has a second thickness that is greater than the previously formed transition metal-metal semiconductor alloy cap portion **27**. During this annealing, the source-side metal semiconductor alloy portion **24s** and the drain-side metal semiconductor alloy portion **24d** are thinned.

The anneal used to provide the thickened source-side transition metal-metal semiconductor alloy portion **26s'**, the thickened drain-side transition metal-metal semiconductor alloy portion **26d'**, and optionally the thickened transition metal-metal semiconductor alloy cap **27'** can be performed at a temperature from 300° C. to 500° C. The anneal used to provide the thickened source-side transition metal-metal semiconductor alloy portion **26s'**, the thickened drain-side transition metal-metal semiconductor alloy portion **26d'**, and optionally the thickened transition metal-metal semiconductor alloy cap **27'** can be performed in an inert ambient such as, for example, helium, argon, neon and/or nitrogen. The anneal used to provide the thickened source-side transition metal-metal semiconductor alloy portion **26s'**, the thickened drain-side transition metal-metal semiconductor alloy portion **26d'**, and optionally the thickened transition metal-metal semiconductor alloy cap **27'** can be performed utilizing one of the anneal processes mentioned above in forming the metal semiconductor alloy portions.

Referring now FIG. 7, there is illustrated the first exemplary semiconductor structure of FIG. 6 after removing remaining portions of the transition metal **26**. The remaining portions of the transition metal **26** can be removed utilizing a selective etching process. As used herein, the term "selective" in reference to a material removal process denotes that the rate of material removal for a first material is greater than the rate of removal for at least another material of the structure to which the material removal process is being applied. In one embodiment, the selectivity may be greater than 10:1. In another embodiment, the selectivity may be greater than 100:1. An example of a selective etch that can be used to remove remaining portions of transition metal **26** comprises sulfuric acid and peroxide mixture OR aqua regia (HNO₃, HCl, and water).

FIG. 7 illustrates one semiconductor structure of the present application. As shown, the semiconductor substrate includes a semiconductor substrate **10** having a source region **22s** and a drain region **22d** located within a semiconductor material portion of the semiconductor substrate **10**, wherein the source region **22s** and the drain region **22d** are spaced apart by a channel region **23**. The semiconductor structure also includes a functional gate structure **18** located above the channel region **15**, a source-side material stack of, from bottom to top, a source-side metal semiconductor alloy portion **24s** and a source-side transition metal-metal semiconductor alloy portion **26s** or **26s'** located on one side of the functional gate structure **18** and located atop the source region **22s**, and a drain-side material stack of, from bottom to top, a drain-side metal semiconductor alloy portion **24d** and a drain-side transition metal-metal semiconductor alloy portion **26d** or **26d'** located on another side of the functional gate structure **18** and located atop the drain region **22d**.

Referring now to FIG. 8, there is illustrated a second exemplary semiconductor structure including at least one sacrificial gate structure **54** formed on an upper surface of a semiconductor substrate **50** in accordance with another embodiment of the present application.

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The semiconductor substrate **50** used in providing the second exemplary semiconductor structure shown in FIG. **8** is the same as semiconductor substrate **10** used in providing the first exemplary semiconductor structure shown in FIG. **1**. Semiconductor substrate **50** may also be processed to include an isolation region **52**. Isolation region **52** can be formed and include materials as described above in forming isolation regions **11** within semiconductor substrate **10**.

The at least one sacrificial gate structure **54** is formed on an upper surface of semiconductor substrate **50**. The term "sacrificial gate structure" is used throughout the present application to denote a material that serves as a placeholder structure for a functional gate structure to be subsequently formed.

The at least one sacrificial gate structure **54** can be formed by first providing a blanket layer of a sacrificial gate material on an upper surface of the semiconductor substrate **50**. The blanket layer of sacrificial gate material can be formed, for example, by chemical vapor deposition or plasma enhanced chemical vapor deposition. The thickness of the blanket layer of sacrificial gate material can be from 50 nm to 300 nm, although lesser and greater thicknesses can also be employed. The blanket layer of sacrificial gate material can include any material that can be selectively removed from the structure during a subsequently performed etching process. In one embodiment, the blanket layer of sacrificial gate material may be composed of polysilicon. In another embodiment of the present application, the blanket layer of sacrificial gate material may be composed of a metal such as, for example, Al, W, or Cu. After providing the blanket layer of sacrificial gate material, the blanket layer of sacrificial gate material can be patterned by lithography and etching so as to form the at least one sacrificial gate structure **54**.

Referring now to FIG. **9**, there is illustrated the second exemplary semiconductor structure of FIG. **8** after forming a spacer **56** on each vertical sidewall of the at least one sacrificial gate structure **54** and forming a source region **58s** on one side of the at least one sacrificial gate structure **54** and a drain region **58d** on another side of the at least one sacrificial gate structure **54**. The semiconductor portion of the semiconductor substrate **50** that is positioned between the source region **58s** and the drain region **58d** is referred as a channel region **59**.

The spacer **56** comprises one of the spacer materials mentioned above in forming spacer **22** to the first exemplary semiconductor structure. Spacer **56** can also be formed utilizing the technique mentioned above in forming spacer **22**. The source region **58s** and the drain region **58d** can be formed by ion implantation as was also mentioned above in forming the source region **24s** and the drain region **24d** in the first exemplary semiconductor structure.

Referring now to FIG. **10**, there is illustrated the second exemplary semiconductor structure of FIG. **9** after forming a dielectric material **60** and thereafter forming a gate cavity **62** by removing the at least one sacrificial gate structure **54**. Dielectric material **60** has an upper surface that is coplanar with an upper surface of the at least one gate structure **54**. As such, the upper surface of the at least one gate structure **54** is exposed after forming the dielectric material **60**.

In some embodiments, the dielectric material **60** may be composed of, for example, silicon dioxide, undoped silicate glass (USG), fluorosilicate glass (FSG), borophosphosilicate glass (BPSG), a spin-on low-k dielectric layer, a chemical vapor deposition (CVD) low-k dielectric layer or any combination thereof. The term "low-k" as used throughout the present application denotes a dielectric material that has a dielectric constant of less than silicon dioxide. In another embodiment, a self-planarizing material such as a spin-on glass (SOG) or a spin-on low-k dielectric material such as

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SiLK can be used as the dielectric material **60**. The use of a self-planarizing dielectric material as dielectric material **60** may avoid the need to perform a subsequent planarizing step.

In one embodiment, the dielectric material **60** can be formed utilizing a deposition process including, for example, chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), evaporation or spin-on coating. In some embodiments, particularly when non-self-planarizing dielectric materials are used as dielectric material **60**, a planarization process or an etch back process follows the deposition of the dielectric material. The thickness of the dielectric material **60** that can be employed in the present application may vary depending on the type of dielectric material employed as well as the method that was employed in forming the same. In one embodiment, the dielectric material **60** has a thickness from 80 nm to 500 nm. Other thicknesses that are greater or lesser than the range provided above can also be used for the dielectric material **60**.

Each gate cavity **62** can be formed by removing each of the sacrificial gate material structures **54** from the structure. Each sacrificial gate material structure **54** can be removed by etching. In one example, a reactive ion etch can be used to removal each sacrificial gate material structure **54**.

Referring now to FIG. **11**, there is illustrated the second exemplary semiconductor structure of FIG. **10** after forming a gate dielectric portion **64** and a gate conductor portion **66** within the gate cavity **62**. The gate dielectric portion **64** and the gate conductor portion **66** provide a functional gate structure **63** within the area previously occupied by the sacrificial gate structure **54**.

In some embodiments and as shown in the drawing, the gate dielectric portion **64** is U-shaped having a bottommost portion in direct contact with an upper surface of the semiconductor substrate **50** and vertical portions that are located on exposed sidewalls of each spacer **56**. Within each gate cavity **62**, the gate dielectric portion **64** surrounds the gate conductor portion **64**. In another embodiment, the gate dielectric portion **64** is not U-shaped and thus lacks the vertical portions mentioned. In such an embodiment, the gate conductor portion **66** that is formed atop the non-U-shaped gate dielectric fills the remaining portion of the gate cavity and has outermost edges that directly contact a sidewall surface of each spacer **56**.

The gate dielectric portion **64** may comprise one of the gate dielectric materials mentioned above for gate dielectric material **14**. Also, the gate dielectric portion **64** may be formed utilizing one of the deposition techniques or thermal growth techniques mentioned above in forming gate dielectric **14**. The thickness of gate dielectric portion **64** is also with the range mentioned above for gate dielectric material **14**.

The gate conductor portion **66** may comprise one of the metal gate conductor materials mentioned above for gate conductor **16**. Also, the gate dielectric portion **66** may be formed utilizing one of the deposition techniques mentioned above in forming gate conductor **16**. The thickness of gate dielectric portion **66** is also with the range mentioned above for gate conductor **16**.

Referring now to FIG. **12**, there is illustrated the second exemplary semiconductor structure of FIG. **11** after forming a dielectric cap material **61** on an upper surface of the dielectric material **60**. The dielectric cap material **61** includes an insulator material that differs from the dielectric material **60**. Examples of insulator materials that can be used as the dielectric cap material **61** include, for example, silicon dioxide, silicon nitride and silicon oxynitride. The dielectric cap material **61** can be formed by a deposition process including, for example, chemical vapor deposition, plasma enhanced

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chemical vapor deposition, evaporation and spin-on coating. The dielectric cap material **61** has a thickness from 1 nm to 20 nm. Other thicknesses can also be used for the thickness of the dielectric cap material **61**.

Referring now to FIG. **13**, there is illustrated the second exemplary semiconductor structure of FIG. **12** after forming a plurality of openings **65** through the dielectric cap material **61** and the dielectric material **60** exposing the source region **58s** and the drain region **58d**. Each remaining portion of the dielectric cap material **60** is referred to herein as dielectric cap material portion **61c**, while each remaining portion of the dielectric material **60** can be referred to herein as dielectric material portion **60'**. The plurality of openings can be formed by lithography and etching. One or more etching processes can be used in forming the plurality of openings **65**. The remaining portion of the dielectric cap portion **61c** is present on a topmost surface of each functional gate structure **63**. As is shown in the drawings of the present application, the dielectric cap portion **61c** that is located on the topmost surface of each functional gate structure **63** extends beyond the vertical edges of the functional gate structure **63** and beyond an outermost edge of each spacer **56**. Also, a portion of each dielectric cap portion **61c** that is present on the topmost surface of the functional gate structure **63** is present on a dielectric material portion **60'** that remains adjacent to the spacer **56**. Further, the dielectric cap portion **61c** and the dielectric material portion **60'** that is present above the isolation regions **52** extend beyond the outer most edges of the isolation regions **52**.

Referring now to FIG. **14**, there is illustrated the second exemplary semiconductor structure of FIG. **13** after forming a source-side metal semiconductor alloy portion **68s** on the source region **58s** and a drain-side metal semiconductor alloy portion **68d** on the drain region **58d**. In this embodiment, no metal semiconductor alloy cap portion is formed atop the gate conductor portion **66**.

The source-side metal semiconductor alloy portion **68s** and the drain-side metal semiconductor alloy portion **68d** can include one of the metal semiconductor alloys mentioned above for source-side metal semiconductor alloy portion **24s** and drain-side metal semiconductor alloy portion **24d**. Further, the source-side metal semiconductor alloy portion **68s** and the drain-side metal semiconductor alloy portion **68d** can be formed utilizing the processing steps mentioned above in forming source-side metal semiconductor alloy portion **24s** and drain-side metal semiconductor alloy portion **24d**.

Referring now to FIG. **15**, there is illustrated the second exemplary semiconductor structure of FIG. **14** after deposition of a transition metal **70** in which a source-side transition metal-metal semiconductor alloy portion **70s** forms preferably in-situ on the source-side metal semiconductor alloy portion **68s** and a drain-side transition metal-metal semiconductor alloy portion **70d** forms preferably in-situ on the drain-side metal semiconductor alloy portion **68d**.

The transition metal **70**, source-side transition metal-metal semiconductor alloy portion **70s** and drain-side transition metal-metal semiconductor alloy portion **70d** are formed as described above in forming transition metal **26**, source-side transition metal-metal semiconductor alloy portion **26s** and drain-side transition metal-metal semiconductor alloy portion **26d**. Also, the transition metal **70**, source-side transition metal-metal semiconductor alloy portion **70s** and drain-side transition metal-metal semiconductor alloy portion **70d** can include one of the transition metal mentioned above in the transition metal **26**, source-side transition metal-metal semiconductor alloy portion **26s** and drain-side transition metal-metal semiconductor alloy portion **26d**. In this embodiment

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of the present application, portions of the transition material **26** are present on the vertical sidewalls of each dielectric material portion **60'** and each dielectric cap material portion **61c**. Also, a portion of the transition metal **26** extends onto an upper surface of each dielectric cap material portion **61c**.

Referring now to FIG. **16**, there is illustrated the second exemplary semiconductor structure of FIG. **15** after annealing. This anneal is optional and thus not be employed in all instances. The anneal step forms a thickened source-side transition metal-metal semiconductor alloy portion **70s'** and a thickened drain-side transition metal-metal semiconductor alloy portion **70d'**. The anneal used in this embodiment of the present application is the same as that described above in forming the first exemplary semiconductor structure shown in FIG. **6**.

Referring now to FIG. **17**, there is illustrated the second exemplary semiconductor structure of FIG. **16** after removing remaining portions of the transition metal **70**. The remaining portions of the transition metal **70** can be removed utilizing a selective etching process such as described above for removing remaining portions of transition metal **26** from the first exemplary semiconductor structure.

FIG. **17** illustrates another semiconductor structure of the present application. As shown the semiconductor substrate includes a semiconductor substrate **50** having a source region **58s** and a drain region **58d** located within a semiconductor material portion of the semiconductor substrate **50**, wherein the source region **58s** and the drain region **58d** are spaced apart by a channel region **59**. The semiconductor structure also includes a functional gate structure **63** located above the channel region **59**, a source-side material stack of, from bottom to top, a source-side metal semiconductor alloy portion **68s** and a source-side transition metal-metal semiconductor alloy portion **70s** or **70s'** located on one side of the functional gate structure **63** and located atop the source region **58s**, and a drain-side material stack of, from bottom to top, a drain-side metal semiconductor alloy portion **68d** and a drain-side transition metal-metal semiconductor alloy portion **70d** or **70d'** located on another side of the functional gate structure **63** and located atop the drain region **58d**.

A conductive material (not shown) can then be deposited filling each opening. The conductive material that is deposited may include for example, at least one of copper, tungsten, and aluminum. In one embodiment, the conductive material that fills each opening comprises copper or a copper alloy such as, for example, a copper-aluminum alloy. The conductive material may be formed by a deposition process including chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), sputtering, or chemical solution deposition. Alternatively, a plating process that fills each contact opening from the bottom upwards can be used. After depositing the conductive material, a planarization process such as, for example, chemical mechanical polishing (CMP) and/or grinding, can be employed to remove portions of the conductive material that extends above the mouth of each opening. In some embodiments, the planarization step can stop on an upper surface of each dielectric cap portion **51c**. In another embodiment, the planarization step can remove each dielectric cap portion **51c** from the structure.

Referring to FIG. **18**, there is illustrated the second exemplary semiconductor structure of FIG. **9** after forming a source-side metal semiconductor alloy portion **68s** on the source region **58s** and the drain-side metal semiconductor alloy portion **68d** of the drain region **58d** in accordance with an alternative embodiment of the present application. In this embodiment, no metal semiconductor alloy cap portion is formed atop the gate conductor portion **66**.

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The source-side metal semiconductor alloy portion **68s** and the drain-side metal semiconductor alloy portion **68d** can include one of the metal semiconductor alloys mentioned above for source-side metal semiconductor alloy portion **24s** and drain-side metal semiconductor alloy portion **24d**. Further, the source-side metal semiconductor alloy portion **68s** and the drain-side metal semiconductor alloy portion **68d** can be formed utilizing the processing steps mentioned above in forming source-side metal semiconductor alloy portion **24s** and drain-side metal semiconductor alloy portion **24d**.

Referring now to FIG. 19, there is illustrated the second exemplary semiconductor structure of FIG. 18 after deposition of a transition metal **70** in which a source-side transition metal-metal semiconductor alloy portion **70s** forms preferably in-situ on the source-side metal semiconductor alloy portion **68s** and a drain-side transition metal-metal semiconductor alloy portion **70d** forms preferably in-situ on the drain-side metal semiconductor alloy portion **68d**.

The transition metal **70**, source-side transition metal-metal semiconductor alloy portion **70s** and drain-side transition metal-metal semiconductor alloy portion **70d** are formed as described above in forming transition metal **26**, source-side transition metal-metal semiconductor alloy portion **26s** and drain-side transition metal-metal semiconductor alloy portion **26d**. Also, the transition metal **70**, source-side transition metal-metal semiconductor alloy portion **70s** and drain-side transition metal-metal semiconductor alloy portion **70d** can include one of the transition metal mentioned above in the transition metal **26**, source-side transition metal-metal semiconductor alloy portion **26s** and drain-side transition metal-metal semiconductor alloy portion **26d**. The source-side transition metal-metal semiconductor alloy portion **70s** and drain-side transition metal-metal semiconductor alloy portion **70d** each contain up to 70% of transition metal.

Referring now to FIG. 20, there is illustrated the second exemplary semiconductor structure of FIG. 19 after annealing. This anneal is optional and thus not need be performed in all instances. The anneal step forms a thickened source-side transition metal-metal semiconductor alloy portion **70s'** and a thickened drain-side transition metal-metal semiconductor alloy portion **70d'**. The anneal used in this embodiment of the present application is the same as that described above in forming the first exemplary semiconductor structure shown in FIG. 6.

Referring now to FIG. 21, there is illustrated the second exemplary semiconductor structure of FIG. 20 after removing remaining portions of the transition metal **70**. The remaining portions of the transition metal **70** can be removed utilizing a selective etching process such as described above for removing remaining portions of transition metal **26** from the first exemplary semiconductor structure.

Referring now to FIG. 22, there is illustrated the second exemplary semiconductor structure of FIG. 21 after forming a dielectric material **60**. The dielectric material **60** that can be employed in this alternative embodiment of the present application includes one of the dielectric materials mentioned above for dielectric material **60**. Also, the dielectric material **60** that can be employed in this alternative embodiment of the present application formed utilizing one of the techniques mentioned above for forming dielectric material **60**.

Referring now to FIG. 22, there is illustrated the second exemplary semiconductor structure of FIG. 21 after forming a gate cavity by removing the at least one sacrificial gate structure **54**, and forming a functional gate structure including a gate dielectric portion **64** and a gate conductor portion **66** within the gate cavity.

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Each gate cavity can be formed utilizing the technique mentioned above in forming the gate cavity in the second exemplary structure of FIG. 10. Also, the description concerning the gate dielectric portion **64** and gate conductor portion **66** used in providing the second exemplary structure shown in FIG. 11 can also be used in providing the gate dielectric portion **64** and gate conductor portion **66** within the gate cavity. In this embodiment, the gate dielectric portion **64** is not U-shaped. Elements **64** and **66** collectively made by referred to as a functional gate structure **63**.

FIG. 22 illustrates another semiconductor structure of the present application. As shown, the semiconductor substrate includes a semiconductor substrate **50** having a source region **58s** and a drain region **58d** located within a semiconductor material portion of the semiconductor substrate **50**, wherein the source region **58s** and the drain region **58d** are spaced apart by a channel region **59**. The semiconductor structure also includes a functional gate structure **63** located above the channel region **59**, a source-side material stack of, from bottom to top, a source-side metal semiconductor alloy portion **68s** and a source-side transition metal-metal semiconductor alloy portion **70s** or **70s'** located on one side of the functional gate structure **63** and located atop the source region **58s**, and a drain-side material stack of, from bottom to top, a drain-side metal semiconductor alloy portion **68d** and a drain-side transition metal-metal semiconductor alloy portion **70d** or **70d'** located on another side of the functional gate structure **63** and located atop the drain region **58d**.

The method of the present application for forming the source-side material stack on a source region and the drain-side material stack on a drain region is not limited to the specific examples mentioned above. Instead, the method of the present application for forming the source-side material stack on a source region and the drain-side material stack on a drain region can be used in any other processing schemes including, for example, formation of FinFET and semiconductor nanowire devices. In addition, the method of the present application can be used in forming contacts to any other type of semiconductor device such as, for example, a hetero-bipolar junction transistor (HBT). Also, the deposition of the transition metal onto a metallic surface of a gate conductor portion may cause some of the transition metal to diffuse into the metallic surface of the gate conductor portion.

While the present application has been particularly shown and described with respect to various embodiments thereof, it will be understood by those skilled in the art that the foregoing and other changes in forms and details may be made without departing from the spirit and scope of the present application. It is therefore intended that the present application not be limited to the exact forms and details described and illustrated, but fall within the scope of the appended claims.

What is claimed is:

1. A semiconductor structure comprising:

- a semiconductor substrate having a source region and a drain region located within a semiconductor material portion of said semiconductor substrate, wherein said source region and said drain region are spaced apart by a channel region;
- a functional gate structure located above said channel region;
- a source-side material stack of, from bottom to top, a source-side metal semiconductor alloy portion and a source-side transition-metal metal semiconductor alloy portion located on one side of said functional gate structure and located atop said source region; and

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a drain-side material stack of, from bottom to top, a drain-side metal semiconductor alloy portion and a drain-side transition metal-metal semiconductor alloy portion located on another side of said functional gate structure and located atop said drain region.

2. The semiconductor structure of claim 1, wherein outermost edges of said source-side metal semiconductor alloy portion and outermost edges of said source-side transition metal-metal semiconductor alloy portion are vertically coincident to each other, and wherein outermost edges of said drain-side metal semiconductor alloy portion and outermost edges of said drain-side transition metal-metal semiconductor alloy portion are vertically coincident to each other.

3. The semiconductor structure of claim 1, wherein said source-side transition metal-metal semiconductor alloy portion is more etch resistant than said source-side metal semiconductor alloy portion, and said drain-side transition metal-metal semiconductor alloy portion is more etch resistant than said drain-side metal semiconductor alloy portion.

4. The semiconductor structure of claim 1, wherein said transition metal of said source-side transition metal-metal semiconductor alloy portion comprises a different metal than said metal of said source-side metal semiconductor alloy portion, and said transition metal of said drain-side transition metal-metal semiconductor alloy portion comprises a different metal than said metal of said drain-side metal semiconductor alloy portion.

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5. The semiconductor structure of claim 4, wherein said semiconductor substrate comprises silicon, said source-side metal semiconductor alloy portion comprises nickel silicide or platinum-nickel silicide, said source-side transition-metal metal semiconductor alloy portion comprises titanium-nickel or titanium-platinum-nickel silicide, said drain-side metal semiconductor alloy portion comprises nickel silicide and said drain-side transition metal-metal semiconductor alloy comprises titanium-nickel silicide or titanium-platinum-nickel silicide.

6. The semiconductor structure of claim 1, wherein said functional gate structure comprises a dielectric material portion and a gate conductor portion, and said dielectric material portion is U-shaped.

7. The semiconductor structure of claim 1, further comprising a dielectric cap material portion located on a topmost surface of said functional gate structure.

8. The semiconductor structure of claim 6, wherein said gate conductor portion includes a transition metal-metal semiconductor alloy cap portion located on a surface of a metal semiconductor alloy cap portion.

9. The semiconductor structure of claim 1, wherein said semiconductor substrate is a III-V compound semiconductor material.

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